learning when exploring available educational opportunities.

VLSI has emerged as a dynamic field offering exciting job prospects for individuals passionate about semiconductor design and integrated circuit technology. In a recent conversation, Kunal Ghosh, VLSI System Design (VSD) founder, shared invaluable insights into the diverse range of job roles within this domain. Unravelling the VLSI job market, Kunal's discussion focused on the multitude of employment opportunities that await individuals interested in VLSI. Kunal discloses, "Most students tend to pursue job opportunities after completing their VLSI education. Some may go for pursuing a master's or PhD degree, while a small percentage may choose to attend different institutes to learn about proprietary tools and practices. However, the majority tends to enter the industry directly after learning VLSI design online courses."

Transitioning from open source to proprietary VLSI tools

The field of VLSI has seen the growing popularity of open source tools for learning and development. However, questions arise regarding the readiness of students who primarily work with open source tools to transition seamlessly to proprietary tools.

- Online tools for VLSI design and development are web-based, accessible from anywhere with an internet connection, and often offer free or subscription-based pricing models.
- Online tools may have a more limited feature set, limited support, and may need better integration with other design tools.
- Intellectual Property (IP) protection can be a concern when using online tools.
- Proprietary tools developed by established companies provide advanced features, extensive libraries, comprehensive functionality, and dedicated technical support.
- Proprietary tools offer better integration with other tools, robust IP protection mechanisms, regular updates, and customisation options.

However, proprietary tools often come with higher costs and require installation on specific systems. The transition from open source to proprietary tools varies among individuals. Learning through open source tools can provide a quick understanding of concepts. The learning curve for proprietary tools is complex and requires extensive brainstorming. Kunal says, "Many students with open source tool experience pick up any proprietary EDA tool, making them more productive, out-of-the-box thinkers, and get hired directly by product and services companies globally. Today, companies are looking for smart students who can figure out things on their own, and open source brings out this skill."

Exploring diverse job profiles in electronic design

The field of electronic design encompasses a wide range of job profiles, each requiring unique skill sets and expertise. From writing register transfer level (RTL) coding to tape-out and tape-in processes, professionals with various specialisations contribute to creating complex integrated circuits. Let's take a closer look at some of these roles:

1. RTL coding (register transfer level coding). One crucial job opening in electronic design involves writing RTL coding. RTL coding serves as the foundation for designing digital circuits, specifying the desired behaviour and functionality of the circuit.

2. Synthesis and gate-level design. After the RTL coding is complete, the synthesis domain is next. Here, the RTL code is converted into gates, creating a gate-level representation of the circuit. This process optimises the circuit's performance, power consumption, and area utilisation.

3. Field programmable gate array (FPGA) design. Field-programmable gate arrays (FPGAs) play a vital role in electronic design. Professionals specialising in FPGA development work on designing and implementing circuits using FPGA technology. They utilise the reconfigurable nature of FPGAs to prototype and test digital circuits rapidly.

4. Analogue design. Analogue design engineers focus on the analogue portion of integrated circuits. They work with IPs and develop circuit designs for functions such as data converters, amplifiers, and filters. Their expertise lies in achieving high-performance analogue circuitry.

5. System-on-chip (SoC) design. SoCs are complex integrated circuits that combine multiple functionalities on a single chip. SoC design engineers are responsible for designing and integrating various IPs and subsystems to create a cohesive and efficient chip. They work on tasks such as bus architecture, interconnect design, and integration of IP blocks.

6. Layout design. Layout engineers collaborate with circuit design engineers to create integrated circuits' physical layouts. They ensure the circuit components are arranged optimally, considering power distribution, signal integrity, and manufacturing constraints.

7. Physical design and SoC implementation. Physical design engineers convert the soft IP into a hard IP. They work on tasks like floor planning, placement, and routing to ensure efficient utilisation of chip area and optimised performance. SoC implementation engineers oversee the entire physical design process and manage the integration of various IP blocks.

8. Physical verification. Physical verification engineers play a critical role in ensuring that the foundry can accurately fabricate the designed circuit. To identify and rectify potential manufacturing issues, they